

Risk assessment as a guide to thrombosis prophylaxis

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Purpose of review

This review discusses the current status of individual thrombosis risk assessment as an alternative to applying prophylaxis to broad groups in an attempt to simplify the process. Complete assessment is necessary in order to provide the proper type, duration, and intensity of prophylaxis, especially following discharge.

Recent findings

The concept of weighting risk factors to obtain a score has been used by some to guide prophylactic decisions for 20 years. This approach was thought to be intuitively correct despite not having been prospectively validated. During the past years, several studies have appeared linking the score to clinically relevant, imaging-proven venous thromboembolism events. In one study, 12.5% of patients with a score higher than 8 developed proven venous thromboembolism events within 60 days postoperatively.

Summary

The use of this score can suggest appropriate prophylaxis for surgical patients, especially those at high risk following hospital discharge.

Keywords

pulmonary embolism, risk assessment, risk factors, thrombosis prophylaxis, venous thrombosis

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Introduction

History and physical examination have long been the standard for providing patient care and used to select therapeutic modalities based on safety and efficacy. Over the past 20 years, sophisticated diagnostic methods have been developed, including ultrasound technology, computerized axial tomography (CT) scans, and MRI. The trend has been to rely more on these exquisite tests than on clinical evaluation. Unfortunately, more limited history and physical examinations have resulted, and assessment of an individual patient for all possible thrombosis risk factors has become unpopular.

During this period, national guidelines have been developed and refined based on the results of randomized clinical trials (RCTs). The results of these analyses are published every few years as a supplement to the journal *CHEST* [1]. They are widely followed and quoted as the Bible for clinical practice decisions, despite the flaw of reflecting only patients that would be included in RCTs. In addition, these guidelines discourage individual risk assessment in the introduction to the chapter on thrombosis prophylaxis, stating that it is cumbersome and not likely to be widely adopted. Several pages later, however, paragraph 1.4.5 reads as follows: ‘The prophylaxis recom-

mendations contained herein are made for groups of patients, for whom the benefits appear to outweigh the risks. However, prophylaxis decisions for an individual patient are best made by combining knowledge of the literature (including the group recommendations provided herein and elsewhere) with clinical judgment (including detailed knowledge of that particular patient’s unique risks for thrombosis, the potential for adverse consequences due to the prophylaxis, and the availability of various prophylaxis options locally). The recommendations that are best for the group may not be best for the individual.’ [1]. In other words, in order to know whether the guidelines apply to an individual patient, there must be an evaluation of that person’s risk factors for thrombosis.

Advice regarding prophylaxis following discharge is limited to suggestions in those having major joint replacement and patients having abdominal cancer operations. Compounding prophylactic issues is the policy being adopted by US regulatory agencies mandating anti-coagulant prophylaxis for most surgical patients unless contraindicated by high bleeding risk. The average length of hospital stay for most of these patients is 2–3 days and there are no data available to show the risk/benefit ratio of 1–3 days of anticoagulant use. Clinical trials used to

develop the *CHEST* guidelines are based on at least 5–7 days of postoperative prophylaxis.

Incidence of venous thromboembolism

It has been estimated that nearly 900 000 venous thromboembolism (VTE) events occur each year in the USA, and this includes 296 000 fatal events primarily from pulmonary emboli. One-third of these fatal events occur in the outpatient setting, which was defined by the authors as occurring more than 90 days following discharge [2]. European investigators have reported very similar statistics indicating that 260 000 hospital-acquired fatal events were seen, along with 108 535 fatalities seen in the community, as reported in 2007 [3]. The *Registro Informatizado de la Enfermedad TromboEmbolica* (RIETE) database contains almost 30 000 patients who have experienced a VTE event from more than 100 worldwide centres. Seventy-seven per cent of these patients suffered their thrombotic event following hospital discharge, 55% of whom suffered the event after anticoagulant prophylaxis had been stopped [4]. The landmark 'Million Women Study' recently published further underscores the late occurrence of VTE in patients having surgery. This study involved 947 454 middle-age female patients who filled out a questionnaire linked to the UK National Health Service hospital admissions and death records. Among these patients, 239 614 had surgery and 5914 were readmitted with VTE, including 270 deaths. Women who underwent a surgical procedure were 70 times more likely to be readmitted within 6 weeks of discharge compared to those not having surgery. Those having day case operations were 10 times more likely to be readmitted than those not having surgery. The risks were lower but still substantially increased 7–12 weeks after surgery [5•].

In another study of 158 325 medical patients discharged with a diagnosis of cancer, congestive heart failure (CHF), lung disease, or severe infectious disease, the mean follow-up was 416–507 days. The incidence of VTE in the cancer patients was 7.6%, which was the highest incidence among all of the groups. Overall, 5.6% of the patients suffered a VTE, which included 1.5% with a pulmonary embolus. These VTE events occurred most frequently following discharge and the median time to occurrence was 74 days. Less than 5% of these patients received postdischarge prophylaxis, although 13.1% of the CHF patients received ongoing prophylaxis. The authors conclude that 'commercially insured medical patients in the US are at high risk of VTE following hospital discharge. One-quarter of medical patients who developed a VTE are at high risk of developing the more severe form of the disease, namely pulmonary emboli, with independent predictors of VTE in the postdischarge period including previous VTE and cancer [6•].'

Risk assessment models

It is well known that not all risk factors carry the same risk of VTE and patients with cancer or a history of previous VTE are much more likely to develop thrombosis compared to aged patients or those at bed-rest. This concept of scoring risk has always made sense but had not been prospectively validated until recently. In a study using electronic alerts, Kucher *et al.* [7] devised a simple score for patients based on eight common risk factors and traced the patients for 90 days after discharge. A score of 4 or more was considered high risk and indeed the authors found a proven 90-day VTE incidence of 5–8% depending upon whether the electronic alerts were followed. The same risk stratification used by Kucher *et al.* was evaluated by studying over 5000 patients in the *Venous thromboembolism RegIsTrY* (VERITY) database in whom the eight factors could be evaluated. They found a linear correlation between the score and proven VTE rate; however, the model lacked low-end sensitivity. Patients with a score of 0–2 had a VTE incidence of more than 20%. This may be explained by noting that the score included only eight factors and likely those with a low score had risk factors not counted in the score [8].

We have designed a risk scoring system taking into account most known risk factors and assigning a score to each factor, which estimates the likelihood of a particular factor being associated with the development of VTE (Fig. 1). The risk list was first evaluated in surgical patients and we were surprised at the number of individuals with risk scores at the highest level [9]. This system was intuitively correct, but despite widespread use over that past two decades, formal validation had not been done. Critics would argue that, as most patients presenting to hospital are 'at risk', a formal and complete assessment is redundant and unnecessary. When one examines this issue more closely, it is apparent that a complete preoperative evaluation is of great value. Patients who are in the highest risk category can be offered combined physical and mechanical prophylaxis during hospitalization, which has been shown to provide superior protection compared to either method alone [1]. A full risk evaluation can provide an accurate patient-informed consent regarding the relative risks and benefits of a surgical procedure, particularly in cases of elective quality-of-life procedures. Some patients after learning their score may decide against cosmetic or other elective procedures. Patients with low scores can be treated with intermittent pneumatic compression (IPC) with good efficacy and no bleeding complications [10]. The most important reason, however, for a thorough risk assessment is to determine the need for continued prophylaxis beyond hospital discharge. The following studies demonstrate the need for providing ongoing prophylaxis in certain patients with high scores.

Figure 1 List of the risk factors and score values used to calculate the overall score and degree of risk

A1: Each risk factor represents 1 point ☐ Age 40–59 years ☐ Minor surgery planned ☐ History of prior major surgery ☐ Varicose veins ☐ History of inflammatory bowel disease ☐ Swollen legs (current) ☐ Obesity (BMI > 30) ☐ Acute myocardial infarction (< 1 month) ☐ Congestive heart failure (< 1 month) ☐ Sepsis (< 1 month) ☐ Serious lung disease incl. pneumonia (< 1 month) ☐ Abnormal pulmonary function (Chronic obstructive pulmonary disease) ☐ Medical patient currently at bed rest ☐ Leg plaster cast or brace ☐ Central venous access ☐ Blood transfusion (< 1 month) ☐ Other risk factor/s_----- A2: For women only (each represents 1 point) ☐ Oral contraceptives or hormone replacement therapy ☐ Pregnancy or postpartum (< 1 month) ☐ History of unexplained stillborn infant, recurrent spontaneous abortion (≥ 3), premature birth with toxemia of pregnancy or growth restricted infant Total risk factor score:	B: Each risk factor represents 2 points ☐ Age 60–74 years ☐ Major surgery (< 60 minutes)* ☐ Arthroscopic surgery (> 60 minutes)* ☐ Laparoscopic surgery (> 60 minutes)* ☐ Previous malignancy ☐ Morbid obesity (BMI > 40) C: Each risk factor represents 3 points ☐ Age 75 years or more ☐ Major surgery lasting 2–3 hours* ☐ BMI > 50 (venous stasis syndrome) ☐ History of SVT, DVT/PE ☐ Family history of DVT/PE ☐ Present cancer or chemotherapy ☐ Present factor V leiden ☐ Positive prothrombin 20210A ☐ Elevated serum homocysteine ☐ Positive lupus anticoagulant ☐ Elevated anticardiolipin antibodies ☐ Heparin-induced thrombocytopenia (HIT) ☐ Other thrombophilia-type_----- D: Each risk factor represents 5 points ☐ Elective major lower extremity arthroplasty ☐ Hip, pelvis or leg fracture (< 1 month) ☐ Stroke (< 1 month) ☐ Multiple trauma (< 1 month) ☐ Acute spinal cord injury (paralysis) (< 1 month) ☐ Major surgery lasting over 3 hours*
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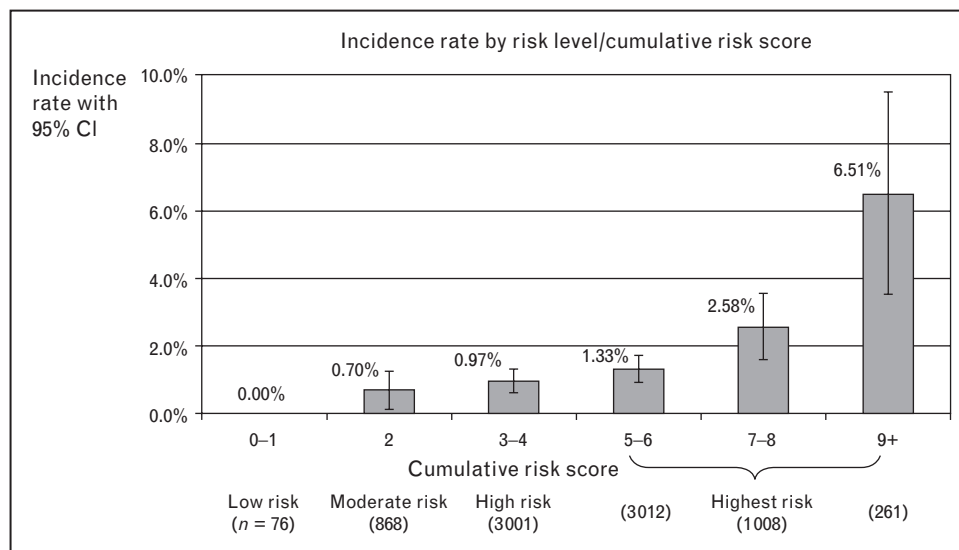
Each risk factor carries a weight depending on the likelihood of that factor causing a thrombotic event. One example would be cancer, which has a score of 3, as it is more likely to result in a thrombosis than bed-rest, which has a score of 1. * Select only one surgical factor from the list.

Seruya *et al.* [11] performed the Caprini risk score in 1156 plastic surgical procedures and found it to be higher than 4 in 173 cases (15%). VTE events were documented in 7.5% (nine patients) and none in the remaining group with lower risk scores. Most of these events occurred in the early postoperative period and 17 of these high-risk patients were sent home on low molecular weight heparin (LMWH) for about 1 week and none developed VTE.

Hatef *et al.* [12] also used the scoring system in 247 plastic surgery patients, 17 of whom developed a deep venous thrombosis (DVT) within 10 days postoperatively. Fifteen of these individuals had a score higher than 4 and the other two a score of 3–4. The authors noted a much higher incidence of DVT (7.7%) in patients who underwent circumferential abdominoplasty compared to 1.7% of the other patients. They proposed that this procedure represents an extra level of risk compared to other plastic surgical procedures.

The largest and most definitive study demonstrating the value of individual patient scores was done by Bahl *et al.* [13**] at the University of Michigan. They conducted a retrospective study involving 8216 surgical inpatients, part of the National Surgical Quality Improvement Program (NSQIP), who were discharged between July 2001 and January 2008. The NSQIP is a well-documented American College of Surgery program that reports 30-day mortality and morbidity outcomes, including the incidence of DVT and pulmonary emboli. The University of Michigan investigators assigned the VTE risk category (low, medium, high, and highest) for each patient based on the Caprini risk assessment model. The risk factors were gathered using data from electronic sources, including hospital billing data, the operating room information system, and clinical data repository. Risk factors for each patient were scored and summed to determine the cumulative VTE risk and associated risk level. The 30-day VTE outcomes were obtained by the

Figure 2 Incidence rate of venous thromboembolism for low-risk, moderate-risk, and high-risk levels and by cumulative risk scores within the highest-risk levels



These thrombotic events represent clinically evident imaging-proven venous thromboembolism (VTE) rates at 30 days postoperatively.

nurses' reviews of patient medical records for evidence of a DVT or pulmonary emboli from diagnostic test results and physician notes. The overall incidence of acquired VTE within 30 days was 1.44%. The incidence of VTE increased with an increase in risk level. VTE was seen in 1.94% of patients in the highest-risk level, 0.97% in the high-risk group, 0.70% in moderate-risk patients, and 0% in low-risk patients. The difference between high-risk and highest-risk levels was statistically significant ($P=0.001$). Further analysis of the highest-risk patients indicated a linear increase in 30-day thrombosis incidence with increasing risk score. Patients with a score higher than 8 had a proven clinical VTE incidence of 6.5% as seen in Fig. 2.

Finally, Pannucci [14] reported the results of a multi-centre University Hospital Project (VTEP) study network involving an analysis of 2000 consecutive plastic surgery patients with a score of more than 3. All patients received pneumatic compression and no anticoagulant prophylaxis postoperatively. Patients were tracked for 60 days and those with a Caprini score higher than 8 had 12.5% clinically evident, imaging-proven VTE events. A second phase of the study is underway involving a similar group, all having LMWH added to the postoperative prophylactic regime.

Conclusion

Individual risk assessment is a valuable tool to estimate the degree of patient risk for VTE and allows prophylaxis

to be tailored for each individual to minimize this dangerous complication. The vast majority of VTE events occur following hospital discharge and after thrombosis prophylaxis has been discontinued. The risk score can suggest ongoing prophylaxis for very high risk individuals who remain 'at risk' for an extended time following surgery.

References and recommended reading

Papers of particular interest, published within the annual period of review, have been highlighted as:

- of special interest
- of outstanding interest

Additional references related to this topic can also be found in the Current World Literature section in this issue (pp. 522–523).

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